DAY 25 | POST-READ

Tools, Graphs and Systems: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Tool calling loop	Describe tools, model requests a call, code validates and runs, result returned, repeat.
Tool schema	name, description, JSON schema parameters, required fields.
Tool safety	Allowlist, validate args, least privilege, human confirmation, step limits.
LangGraph	StateGraph(State), add_node, add_edge, add_conditional_edges, compile, invoke.
Cycles	Retries and agent loops with a counter in state to stop them.
Iceberg model	Events, patterns, structures, mental models.
Feedback loops	Reinforcing grows, balancing stabilises. Delays cause overshoot.
Systems traps	Fixes that fail, shifting the burden, ignoring delays.

Part B: Going beyond today

1. Parallel tool calls

Models can request several tool calls in one turn (for example, two order lookups). Run independent calls in parallel and return all results before the next model call.

2. Human in the loop with checkpoints

```
from langgraph.checkpoint.memory import MemorySaver
app = graph.compile(checkpointer=MemorySaver(), interrupt_before=["issue_refund"])
```

The graph pauses before the risky node; a person reviews the state and resumes or cancels.

3. Tool descriptions are prompts

Most wrong tool choices come from vague names and descriptions. Write them as if explaining to a new colleague, including an example argument.

4. Leverage points

In systems, small changes in the right place (information flows, rules, goals) often beat big pushes in the wrong place (adding more staff to a broken process).

5. Simulate before you commit

A tiny simulation, like today's backlog model, will not predict the future precisely, but it reveals the shape of the behaviour: delays, overshoot and tipping points.

Part C: Check yourself

- Who executes a tool, the model or your code?
- Name three tool safety rules.
- What is the difference between an edge and a conditional edge?
- Give an example of a reinforcing loop in a customer support team.
- What is a leverage point?

Part D: Practice ideas for the weekend

- Add a create_ticket tool that requires human confirmation before it runs.
- Add a human approval interrupt to the LangGraph workflow.
- Draw the iceberg model for a recurring problem in your own work.

Coming up on Day 26: agentic AI and agentic RAG: agent components, the OpenAI Agents SDK, corrective RAG and multi-agent LangGraph.

Yoi shuumatsu wo (良い週末を): have a great weekend